

Inclusion–Exclusion

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COMP2711 Discrete Mathematical Tools

1 Counting overlapping sets

1.1 Two sets

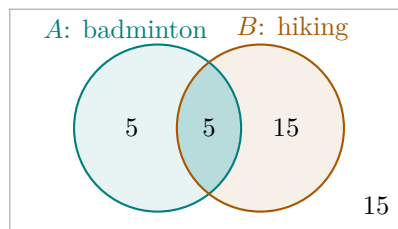
Among 40 classmates, 10 play badminton and 20 go hiking. How many do neither activity? Is the answer simply $40 - 10 - 20 = 10$?

Not necessarily. Bob may play badminton *and* go hiking, so he appears in both counts. We cannot determine the answer until we know how many students do both.

Now suppose 5 students do both. Let A be the badminton group and B the hiking group. Adding $10 + 20$ counts these 5 students twice. Subtracting the overlap once gives

$$|A \cup B| = 10 + 20 - 5 = 25.$$

Therefore $40 - 25 = 15$ classmates do neither activity.



Must understand: Inclusion–exclusion

For two finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Add the two counts, then subtract their overlap once. If the sets are disjoint, the overlap is empty and this becomes the ordinary sum rule.

We will first extend this correction to more sets. Then we will use it to count valid outcomes: count all candidates and subtract those that break at least one rule.

1.2 Three sets

A new survey of 40 students records three activities:

- 22 play badminton, 18 go hiking, and 15 play board games.
- 8 do badminton and hiking; 7 do badminton and board games; 6 do hiking and board games.
- 3 do all three activities.

How many do at least one activity? How many do none?

Let A, B, C denote the badminton, hiking, and board-game groups, respectively. The survey gives

$$\begin{aligned} |A| &= 22, & |B| &= 18, & |C| &= 15, \\ |A \cap B| &= 8, & |A \cap C| &= 7, & |B \cap C| &= 6, \\ |A \cap B \cap C| &= 3. \end{aligned}$$

Each pairwise intersection *includes* the students who do all three. For example, the 6 students who do hiking and board games may also play badminton.

Solution 1: Fill the Venn diagram. Start with the most constrained region, then work outward.

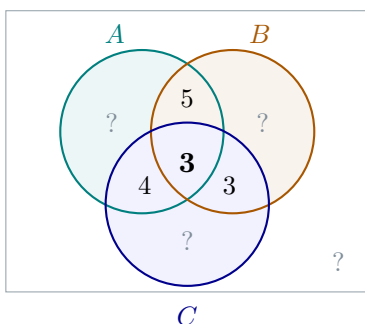
Step 1: All three. Put 3 in the center, $A \cap B \cap C$.

Step 2: Exactly two. The 6 students in $B \cap C$ include the 3 in the center. Thus the region for hiking and board games *but not badminton* has $6 - 3 = 3$ students. The other pair-only regions follow in the same way:

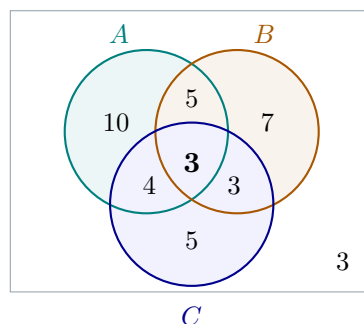
$$\begin{aligned} \text{Badminton and hiking only: } & 8 - 3 = 5, \\ \text{Badminton and board games only: } & 7 - 3 = 4, \\ \text{Hiking and board games only: } & 6 - 3 = 3. \end{aligned}$$

Step 3: Exactly one. Subtract the already filled regions of each circle from its total:

$$\begin{aligned} \text{Badminton only: } & 22 - 5 - 4 - 3 = 10, \\ \text{Hiking only: } & 18 - 5 - 3 - 3 = 7, \\ \text{Board games only: } & 15 - 4 - 3 - 3 = 5. \end{aligned}$$



Center and pair-only regions.



The completed diagram.

The seven regions inside the circles are disjoint, so we can add their counts:

$$|A \cup B \cup C| = 10 + 7 + 5 + 5 + 4 + 3 + 3 = 37.$$

Thus $40 - 37 = 3$ students do none. Exactly two activities means one of the three pair-only regions, giving $5 + 4 + 3 = 12$ students.

Solution 2: Inclusion–exclusion. We can obtain the same union count without filling every region. For three finite sets,

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

Why add the triple intersection? A student in all three groups is added three times and subtracted three times. Adding the triple intersection counts that student once, as required.

Applying the formula gives

$$22 + 18 + 15 - 8 - 7 - 6 + 3 = 37.$$

This agrees with the completed diagram: 37 do at least one activity, and 3 do none.

1.3 The general principle

For finite sets $A_1, \dots, A_n$, where $n \geq 1$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|.$$

Add all single-set counts, subtract all two-set intersection counts, add all three-set intersection counts, and continue with alternating signs. Count each choice of sets once; for example, $i < j$ avoids listing both $A_i \cap A_j$ and $A_j \cap A_i$.

Proof. Check how often one element is counted. If an element belongs to exactly $r \geq 1$ sets, it occurs in $\binom{r}{1}$ single sets, $\binom{r}{2}$ pairwise intersections, and so on. Its total contribution is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \dots + (-1)^{r+1} \binom{r}{r} = 1,$$

because the binomial theorem gives

$$0 = (1 - 1)^r = 1 - \binom{r}{1} + \binom{r}{2} - \dots + (-1)^r \binom{r}{r}.$$

An element outside every set contributes zero. Thus the formula counts exactly the elements in the union, each once.

1.4 When to use inclusion–exclusion

Case 1: At least one condition. Count outcomes satisfying condition A or B or C , and so on. The OR count may be difficult because the conditions overlap. Count specified conditions together—the AND cases—and apply inclusion–exclusion. For example, divisibility by 2 and by 3 is just divisibility by 6.

Case 2: None of the conditions. Start with a finite universe U of candidates. If A_i is the set of outcomes satisfying bad condition i , a valid outcome satisfies none of those conditions. Thus

$$\#\text{valid} = |U| - \left| \bigcup_{i=1}^n A_i \right| = |U| - \sum_i |A_i| + \sum_{i < j} |A_i \cap A_j| - \dots.$$

Every intersection must count outcomes from the same universe.

2 Applications

Application 1: Divisibility

How many integers from 1 through 100 are divisible by at least one of 2, 3, 5? Let A_d be the multiples of d in this range.

The multiples of a positive integer d are $d, 2d, 3d, \dots$. The last multiplier is $\lfloor 100/d \rfloor$, where $\lfloor x \rfloor$ is the greatest integer not exceeding x . Therefore $|A_d| = \lfloor 100/d \rfloor$.

A number lies in $A_a \cap A_b$ exactly when it is divisible by both a and b , or equivalently, by their least common multiple $\text{lcm}(a, b)$.

Set	Its elements are multiples of	Count
A_2, A_3, A_5	2, 3, 5	50, 33, 20
$A_2 \cap A_3$	6	16
$A_2 \cap A_5$	10	10
$A_3 \cap A_5$	15	6
$A_2 \cap A_3 \cap A_5$	30	3

Thus

$$|A_2 \cup A_3 \cup A_5| = 50 + 33 + 20 - 16 - 10 - 6 + 3 = 74.$$

There are $100 - 74 = 26$ integers divisible by none of 2, 3, 5.

Follow-up. How many integers from 1 through 100 are divisible by 6 or 10? The common multiples are multiples of $\text{lcm}(6, 10) = 30$, not $6 \cdot 10 = 60$. The answer is

$$\left\lfloor \frac{100}{6} \right\rfloor + \left\lfloor \frac{100}{10} \right\rfloor - \left\lfloor \frac{100}{30} \right\rfloor = 16 + 10 - 3 = 23.$$

Application 2: Digit codes

How many five-character codes use only 0, 1, 2 and contain each digit at least once? Repetition and leading zeros are allowed. For example, 00122 is valid, but 00110 is not: digit 2 is missing.

A direct attempt is to choose one position for each required digit and then fill the remaining positions. This overcounts unevenly. The final code 00122 has $2 \cdot 1 \cdot 2 = 4$ ways to designate the required occurrences, whereas 00012 has $3 \cdot 1 \cdot 1 = 3$. There is no single repetition factor by which to divide. Instead, count codes that miss a digit.

1. **All candidates.** Let U contain all five-character codes over $\{0, 1, 2\}$. There are $|U| = 3^5$ codes.
2. **Bad sets.** Let E_i contain the codes missing digit i , for $i \in \{0, 1, 2\}$.
3. **Intersections.** A code in an intersection must avoid every specified digit:

Forbidden digits	Choices per position	Number of codes
One specified digit is missing	2	2^5
Two specified digits are missing	1	1^5
All three digits are missing	0	0

There are three choices of one missing digit and $\binom{3}{2} = 3$ choices of a pair.

4. **Inclusion–exclusion.** The required count is

$$3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150.$$

For example, the code 00000 belongs to both E_1 and E_2 . Subtracting each bad set removes it twice; adding their intersection corrects this.

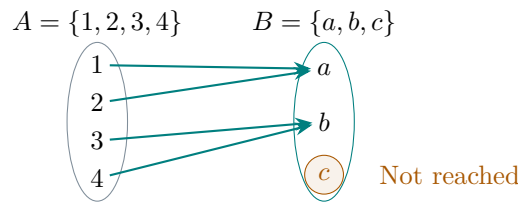
Application 3: Surjective functions

Let $|A| = m \geq 1$ and $|B| = n \geq 1$. Each input can independently map to any of the n outputs, so there are n^m functions $f : A \rightarrow B$.

A function is **surjective** if every element of B is reached. For $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$, the function

$$f(1) = f(2) = a, \quad f(3) = b, \quad f(4) = c$$

is surjective. By contrast, the function below misses c .



A function fails to be surjective if it misses b_1 or b_2 or $\dots$ or b_n . Use the complement case: count all functions, then remove those missing at least one output.

If k specified outputs are forbidden together, every input has $n - k$ choices, giving $(n - k)^m$ functions. There are $\binom{n}{k}$ choices of those forbidden outputs. Hence

$$\#\text{surjective functions} = n^m - \binom{n}{1}(n - 1)^m + \binom{n}{2}(n - 2)^m - \dots = \sum_{k=0}^n (-1)^k \binom{n}{k} (n - k)^m.$$

The $k = 0$ term counts all functions. A term counts functions missing *at least* the specified outputs; other outputs may also be missing. The alternating corrections handle that overlap. If $m < n$, the answer is zero: m inputs cannot reach n outputs.

Application 4: Derangements

Permute the numbers $1, 2, \dots, n$. How many permutations have no number i in position i ? Such a permutation is a **derangement**.

For $n = 4$, $(2, 3, 4, 1)$ is valid. The permutation $(2, 3, 1, 4)$ is invalid: number 4 remains in position 4.

Position	1	2	3	4	1	2	3	4
	2	3	4	1	2	3	1	4
	Valid				Invalid: 4 is in position 4			

A valid permutation has 1 not in position 1 *and* 2 not in position 2, and so on. Its complement has 1 in position 1 *or* 2 in position 2, and so on. Count the invalid permutations by inclusion–exclusion.

There are $n!$ candidates. If k specified numbers stay in their own positions, the remaining numbers can be permuted in $(n - k)!$ ways. Choose those k numbers in $\binom{n}{k}$ ways. Thus the number D_n of derangements is

$$D_n = \sum_{k=0}^n (-1)^k \binom{n}{k} (n - k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

The second equality uses $\binom{n}{k}(n - k)! = n!/k!$. The other numbers may also be in their own positions while counting an intersection; inclusion–exclusion removes all such invalid outcomes in the final sum. For four numbers,

$$D_4 = 4! - \binom{4}{1}3! + \binom{4}{2}2! - \binom{4}{3}1! + \binom{4}{4}0! = 24 - 24 + 12 - 4 + 1 = 9.$$

We use $D_0 = 1$ for the empty arrangement and $D_1 = 0$.

Highly recommended challenge: No pair swaps. How many derangements of $1, 2, \dots, n$ also have no two numbers swapping their positions? For example, $(2, 1, 5, 3, 4)$ is a derangement, but is invalid under the additional rule: 1 and 2 swap positions. Use inclusion–exclusion.

Hint. Define a bad event for each fixed point and for each pair swap. Which collections of events can hold simultaneously?

Solution. Let $\pi(i)$ be the number in position i in a permutation of $\{1, \dots, n\}$. Define

$$A_i = \{\pi : \pi(i) = i\}, \quad B_{ij} = \{\pi : \pi(i) = j, \pi(j) = i\} \quad (1 \leq i < j \leq n).$$

We must avoid every A_i and every B_{ij} . Two distinct selected bad events cannot share a position: $A_1 \cap B_{12} = \emptyset$ and $B_{12} \cap B_{23} = \emptyset$, for example. The first would require both $\pi(1) = 1$ and $\pi(1) = 2$; the second would require both $\pi(2) = 1$ and $\pi(2) = 3$. Thus every nonempty intersection specifies a fixed points and b disjoint pair swaps, using $a + 2b \leq n$ positions in total.

Choose the fixed positions, then the positions in pair swaps, then pair them:

$$\binom{n}{a} \binom{n-a}{2b} \frac{(2b)!}{2^b b!}.$$

For the last factor, order the $2b$ selected positions and group consecutive positions into pairs. Each pairing appears $2^b b!$ times: either order within each pair, and any order of the b pairs.

Once these bad events are forced, permute the remaining $n - a - 2b$ numbers among the unused positions. They may have additional fixed points or pair swaps; an intersection requires the selected bad events, not exactly those events. For this choice of a, b , the total intersection count is

$$\binom{n}{a} \binom{n-a}{2b} \frac{(2b)!}{2^b b!} (n-a-2b)! = \frac{n!}{a! 2^b b!}.$$

There are $a + b$ selected bad events, so the inclusion–exclusion sign is $(-1)^{a+b}$, not $(-1)^{a+2b}$. Including $a = b = 0$ for all $n!$ candidates, the answer is

$$\sum_{\substack{a, b \geq 0 \\ a+2b \leq n}} (-1)^{a+b} \frac{n!}{a! 2^b b!}.$$

Checks using cycles. Draw an arrow from i to $\pi(i)$. Each vertex has one incoming and one outgoing arrow, so a permutation is a collection of disjoint directed cycles. A fixed point is a cycle of length 1; a pair swap is a cycle of length 2. Our condition therefore requires every cycle to have length at least 3. For $n = 4$, only a single 4-cycle is possible, giving $(4 - 1)! = 6$ permutations. For $n = 6$, the possibilities are a single 6-cycle or two 3-cycles:

$$5! + \frac{\binom{6}{3}}{2!} (3 - 1)!^2 = 120 + 40 = 160.$$

For a cycle on k chosen numbers, fix one number as the starting point and order the other $k - 1$, giving $(k - 1)!$ cycles. For the two 3-cycles, divide by $2!$ because choosing either triple first gives the same pair of groups. Both values agree with the inclusion–exclusion sum.

Application 5: Integer solutions

Let n, r be nonnegative integers. How many integer triples satisfy

$$x_1 + x_2 + x_3 = n, \quad 0 \leq x_1, x_2, x_3 \leq r?$$

For example, when $n = 8$ and $r = 3$, $(3, 3, 2)$ is valid, but $(4, 3, 1)$ violates the upper bound on x_1 .

Ignore the upper bounds first. Stars and bars counts all nonnegative solutions to $x_1 + x_2 + x_3 = n$: there are n stars and two bars, giving $\binom{n+2}{2}$ candidates. For example, the two bars in

$$\underbrace{\star\star\star}_{x_1=3} \mid \underbrace{\star\star\star}_{x_2=3} \mid \underbrace{\star\star}_{x_3=2}$$

separate the eight stars into the groups for $(3, 3, 2)$.

Remove the invalid solutions. A solution is invalid when

$$x_1 \geq r + 1 \quad \text{or} \quad x_2 \geq r + 1 \quad \text{or} \quad x_3 \geq r + 1.$$

Use inclusion–exclusion to count these overlapping conditions.

For example, an intersection requires $x_1 \geq r + 1$ and $x_2 \geq r + 1$, while $x_1 + x_2 + x_3 = n$. Subtract $r + 1$ from each of the first two variables:

$$\underbrace{x_1 - (r + 1)}_{\geq 0} + \underbrace{x_2 - (r + 1)}_{\geq 0} + \underbrace{x_3}_{\geq 0} = n - 2(r + 1).$$

This is ordinary stars and bars with a smaller total, giving

$$\binom{n - 2(r + 1) + 2}{2} = \binom{n - 2r}{2}.$$

Adding $r + 1$ back to the first two entries recovers the original solution. For one chosen variable, subtract $r + 1$ once; for all three, subtract it three times. Only the chosen variables are required to exceed r ; the others may exceed it too.

If the remaining total is negative, the count is zero. Use $\binom{a}{2} = 0$ for an integer $a < 2$ in the table and formula below.

Require	Count per choice	Choices
One chosen variable $\geq r + 1$	$\binom{n-r+1}{2}$	3
Two chosen variables $\geq r + 1$	$\binom{n-2r}{2}$	3
All three variables $\geq r + 1$	$\binom{n-3r-1}{2}$	1

Apply inclusion–exclusion. There are three choices of a single variable, three pairs, and one triple. The valid count is all candidates minus the singles, plus the pairs, minus the triple:

$$\binom{n+2}{2} - 3\binom{n-r+1}{2} + 3\binom{n-2r}{2} - \binom{n-3r-1}{2}.$$

For $n = 8, r = 3$, a single violation leaves a total of four, a pair leaves zero, and three violations would require twelve units. Thus

$$\binom{10}{2} - 3\binom{6}{2} + 3\binom{2}{2} - 0 = 45 - 45 + 3 = 3.$$

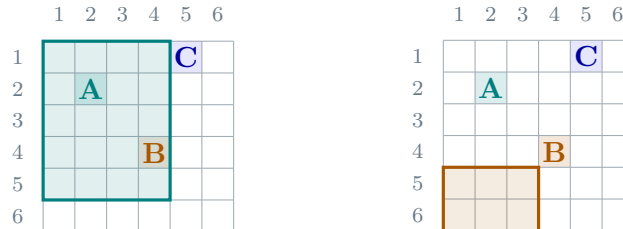
The three solutions are $(2, 3, 3)$, $(3, 2, 3)$, and $(3, 3, 2)$.

Application 6: Rectangles containing marked cells

Mark cells A, B, C in an $n \times n$ grid. How many rectangles contain at least one whole marked cell? Rectangles have positive area, stay inside the grid, and have edges on grid lines.

We use the 6×6 example below, numbering rows from top to bottom and columns from left to right:

$$A = (2, 2), \quad B = (4, 4), \quad C = (1, 5).$$



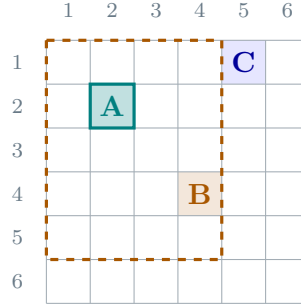
Valid: contains A and B . Invalid: contains no marked cell.

Let R_A, R_B, R_C be the sets of rectangles containing A, B, C , respectively. The target is $|R_A \cup R_B \cup R_C|$. These are sets of *rectangles*, not sets of grid cells. We first count the three single sets, then the pairwise and triple intersections, and combine them using inclusion–exclusion.

Choose four boundaries. A rectangle is uniquely determined by its top, bottom, left, and right edges. To contain the cell in row r , column c , its top edge has r choices and its bottom edge has $n - r + 1$ choices. Similarly its left edge has c choices and its right edge has $n - c + 1$ choices. These choices are independent and always enclose the cell, so their product is

$$r(n - r + 1)c(n - c + 1).$$

There is no division: every rectangle determines exactly one choice of these four named boundaries.



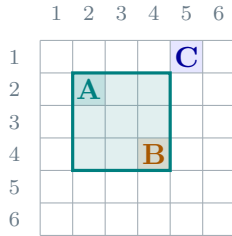
The solid outline encloses A alone; the dashed outline is one larger rectangle containing A . Rectangles counted in R_A may also contain B or C . For A , the four boundary counts are 2, 5, 2, 5. Thus the three single-set counts are

$$|R_A| = 2 \cdot 5 \cdot 2 \cdot 5 = 100, \quad |R_B| = 4 \cdot 3 \cdot 4 \cdot 3 = 144, \quad |R_C| = 1 \cdot 6 \cdot 5 \cdot 2 = 60.$$

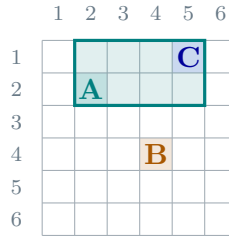
Contain the bounding box. To contain several specified cells, a rectangle must contain their smallest enclosing rectangle, or *bounding box*. Conversely, every rectangle containing that box contains all the specified cells. We can therefore use the same four-boundary count.

For example, the bounding box of A and B spans rows 2 through 4 and columns 2 through 4. A containing rectangle has 2 choices for its top edge, 3 for its bottom edge, 2 for its left edge, and 3 for its right edge:

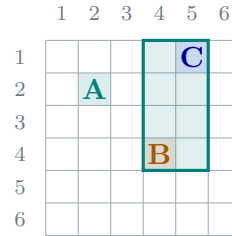
$$|R_A \cap R_B| = 2 \cdot 3 \cdot 2 \cdot 3 = 36.$$



$$A, B: \text{ rows 2-4, cols. 2-4} \\ 2 \cdot 3 \cdot 2 \cdot 3 = 36$$

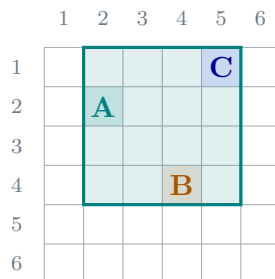


$$A, C: \text{ rows 1-2, cols. 2-5} \\ 1 \cdot 5 \cdot 2 \cdot 2 = 20$$



$$B, C: \text{ rows 1-4, cols. 4-5} \\ 1 \cdot 3 \cdot 4 \cdot 2 = 24$$

A pair count does not exclude the third marked cell.



For all three cells, the bounding box spans rows 1 through 4 and columns 2 through 5. Hence

$$|R_A \cap R_B \cap R_C| = 1 \cdot 3 \cdot 2 \cdot 2 = 12.$$

Inclusion–exclusion now gives

$$\begin{aligned} |R_A \cup R_B \cup R_C| &= 100 + 144 + 60 - 36 - 20 - 24 + 12 \\ &= 236. \end{aligned}$$

Thus exactly 236 rectangles contain at least one marked cell in the displayed grid.

Arbitrary positions in a square grid. For a nonempty selection S of marked cells, let $r_{\min}, r_{\max}$ be its smallest and largest row indices, and $c_{\min}, c_{\max}$ its smallest and largest column indices. These specify its bounding box. The number of rectangles containing every cell in S is

$$F(S) = r_{\min}(n - r_{\max} + 1)c_{\min}(n - c_{\max} + 1).$$

The four factors count the top, bottom, left, and right boundary choices, in that order. For three marked cells the answer is

$$F(\{A\}) + F(\{B\}) + F(\{C\}) - F(\{A, B\}) - F(\{A, C\}) - F(\{B, C\}) + F(\{A, B, C\}).$$

The count depends on the marked positions, not only on n .

3 Practice

1. Among 200 students, 120 study mathematics, 90 study computing, and 50 study both. How many study at least one subject? How many study neither?
2. How many integers from 1 through 500 are divisible by 6 or 10?
3. Find D_5 .
4. In the displayed 6×6 grid, how many rectangles contain none of A, B, C ?
5. Three clubs have 20, 18, and 16 members. Their pairwise intersection sizes are 7, 6, and 5, and 39 people belong to at least one club. How many belong to all three?

Answers

1. At least one: $120 + 90 - 50 = 160$. Neither: $200 - 160 = 40$.
2. The common multiples are multiples of 30. The answer is $\lfloor 500/6 \rfloor + \lfloor 500/10 \rfloor - \lfloor 500/30 \rfloor = 83 + 50 - 16 = 117$.
3. $D_5 = 120 - 120 + 60 - 20 + 5 - 1 = 44$.
4. Choose two of the seven horizontal grid lines and two of the seven vertical grid lines. There are $\binom{7}{2}^2 = 441$ rectangles in total, so $441 - 236 = 205$ contain none of the marked cells.
5. Let t be the triple-intersection size. Inclusion-exclusion gives $39 = 20 + 18 + 16 - 7 - 6 - 5 + t = 36 + t$, so $t = 3$.

Main ideas

- Add simple counts and correct repeated counts by alternating intersections.
- To satisfy every restriction, remove outcomes that violate at least one restriction.
- Define the universe and interpret the intersections before choosing a formula.